QC LAB DIGITIZATION (QCLAD)_WORKSHOP

ANSON ALEXANDER ,MELVIN ANG, NEELIMA RAJESH

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QCLAD PROJECT CHARTER

Project Background / Rationale

- *Opportunity to improve the efficiency of end-to-end QC lab management to significantly reduce the non-value added time in manual tasks, planning and execution so that QC team can focus on more value-added activities.*
- *Limited visibility and standardization of QC lab best practices across regions and globally.*
- *Opportunity to digitalize information flow and acquire new data to enable advanced analytics and machine learning use cases towards a predictive quality future state.*

Project Objectives

- *To digitalize QC lab operations to improve labor productivity, information flow and transparency.*
- *Reduce majority of QC time spend on manual processes, documentations & operations planning.*
- *Pilot 1 site in 2024 (Dragon)*
- *Blue Print for further APAC deployments and scaleup's further (10 sites) in 2025*

Project Deliverables

- *QCLAD deployed at Dragon, phase A by Dec24' and phase B by Mar25'.*
- *Integration completed between SAP, QCLAD, MES , test instruments (as per SoW) and Skills Matrix (OSDP) to realize automatic QC data capture and transfers.*
- *Deployment of capabilities as per project scope in SoW (refer slide 3)*
- *Successful project management and pilot delivery to enable future scalability*

CAPEX Budget for Dragon Pilot

Vendor Budget	**392T€**

Key Project Dates

Business case and technical requirements	Jan 2024
Buget approval and PO creation	Apr - May 2024
Process and equipment standardization	Apr - Aug 2024
Pilot launch Workshop@Dragon	Jun 2024
Sign off of Functional Requirement Specifications	Jul 2024
Dragon pilot site deployment (phase A+B)	Jun 2024 - Mar 2025
Key User Trainings	Aug -Sep 2024
Create blueprint for APAC rollout	Nov 2024
User feedback / enhancements review post deployment	Apr 2025

QCLAD - PROJECT SCOPE
HIGH-LEVEL CONCEPT / FUNCTIONAL SOLUTION SCOPE

Sample Management
- RM, Bulk, FG sample whole life cycle management
- Sample receipt
- Sample testing/results entry
- Sample results approve/reporting
- Special template CoA creation

Sampling Scheduler
- Pre-task/test scheduled before sample arrived
- Manual/Automatic task/resource arrangement

Non-Routine Test Management
- Test requirement register
- Test conduct/results entry
- Test results approve/reporting

RM CoA Capturing (with eCOA integration in future)
- Capture data from RM CoA
- Upload vendor test data to SAP
- Email Warning if abnormality

Not in Scope for Dragon Pilot

* Henkel eCOA solution integration is future enhancement

Lab Execution System
- Daily test operation workflow execution
- QC Lab SOP/test method execution
- Calculation & data processing
- Enable eSignatures & cross checking between users
- Generate follow-up actions

QCLaD
QC Lab Digitization

Interface with Other Henkel systems
- SAP QM /MES/OSDP(Skill Matrix)
- Data communication between different systems (Material master data, e.g., IDH#, shelf life, Batch#, etc. Inspection plan/lot/result/MIC#, instrument info, Personnel skill matrix, etc.)

Lab Management
- Resource & Inventory Mgt
- Instrument (Instrument Connectivity, Maintenance, Calibration, etc.)
- Consumables (Chemical, Reagent, Standard, etc.)
- Personnel

Data Capturing from Instrument

Reporting & Dashboard
- Sample status summary
- Instrument load, sample workload statistics
- KPI monitoring

Security & User Access Management
- Audit trails
- User account, access, etc.

Mobile Device (Tablet)

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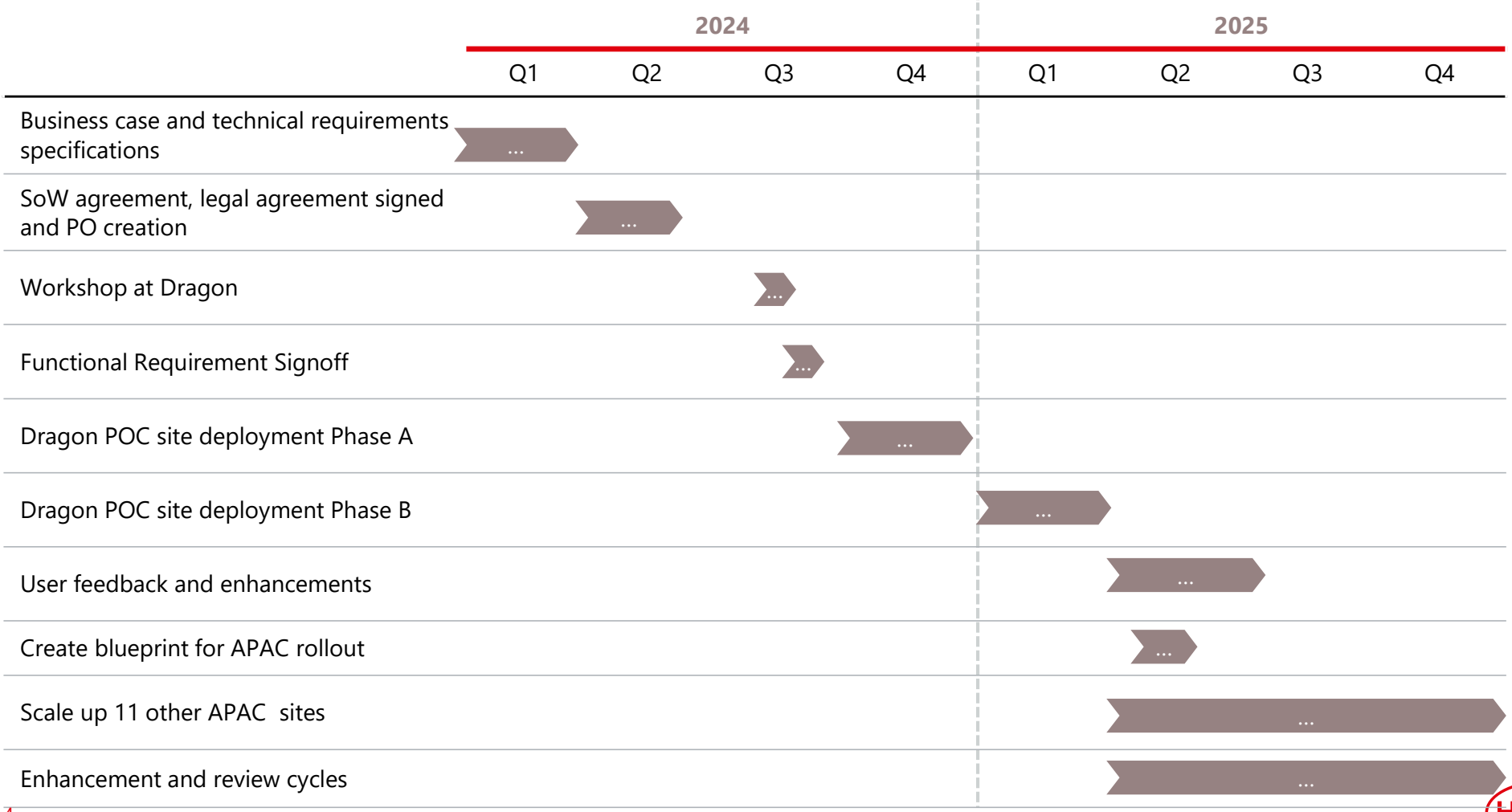
TIMELINE AND KEY MILESTONES



WORKSHOP DELIVERABLE & EXPECTATIONS _HENKEL

To Update

- Henkel process and User requirement specification will be completely understood by ThermoFisher
- Gaps with capabilities from URS. Label printing done in QCLAD – what is expected from Henkel?
- Roles and responsibilities on Henkel and ThermoFisher
- Detailing expectations and deliverables from Henkel and 3rd party application vendors(API calls..) for seamless execution and an estimated timeline when it should be ready to meet Phase A and Phase B (Infrastructure, Integration – SAP, MES, Skill matrix) timelines.
- Detailed steps and timeline for each milestone - what can be expected, when can be expected, dependencies, which team are required..
- Consensus on working pattern
  - meeting frequency, participation plan (for resource availability)
  - mode of operation (Henkel Devops is recommended) for all communication (user stories, product backlog, sprint planning, release), responsibilities in the operation
  - escalation process

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WORKSHOP DELIVERABLE & EXPECTATIONS_THEMOFISHER

To Update

- TO BE UPDATED

WORKSHOP DELIVERABLES (JUN 3-7)

To Update

Day 1	Day 2	Day 3	Day 4	Day 5

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PROJECT ORGANIZATION_HENKEL

Draft

Steering Committee
- Bjoern Jackisch
- MarieChristine Martinez
- Nick Meisen
- Jaume Cornellana

3rd party vendors
Savita (Vegam)
Arun Gopal (Skill Matrix)

Project Management Office (PMO)
- Anson Alexander
- Maline Sun
- Alvin Xie
- Melvin Ang
- Szymon Bliski

Product Team
Neelima Rajesh (Global)
Joanna (Global)

Project Management
Frank SHEN (Dragon)
David Li (APAC)
Samantha Zhang (Dragon)

Project Core Team

Workstream 1
Quality Team
- Lisa Zhou
- Tony Zhuang
- Joanna Wang

Workstream 2
Dx Team
- Yenli Khor
- Mirko Feisel
- Ashu Singh
- TBD

Workstream 3
CoE
- Alexis
- Katja

Extended Team

SME's
- Chaihui Chong

SME's
- John Riordan
- Priyanka Wani
- Sreeraj Devaraj
- Khun Zhang
- Avinash Nirgude

SME's

Steering Bodies
- Decides key issues
- Ensures adequate allocation of resources
- Frequency: To be defined, vs. fortnight

Project Management Office
- Support of the project management and the project team in recording and information tasks

Project Manager
- Responsible for overall management and results

Project Core Team
- Operational control of the work streams
- Daily project work and QM
- Secures project progress
- Obtains and documents results from defined workstreams

Extended Team
- Expert knowledge to support core team members in specific task
- Participation on request

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PROJECT ORGANIZATION_THERMOFISHER

Draft

QC LAB DIGITALIZATION (QCLAD) BUSINESS CASE

MELVIN ANG, AOD,

ANSON ALEXANDER, QUALITY APAC

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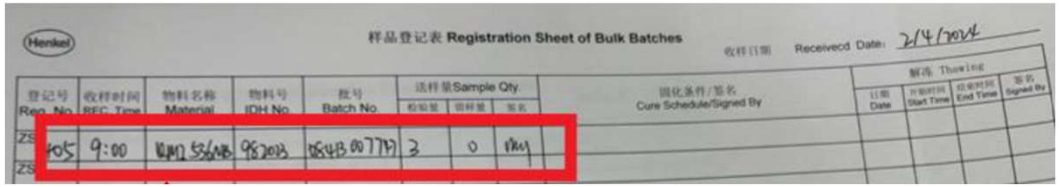
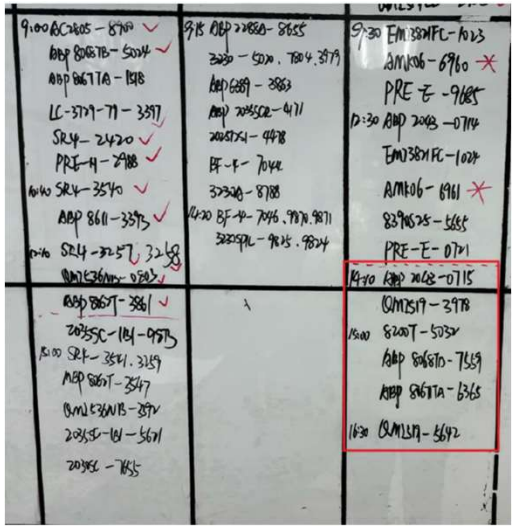
QC LAB DAILY TESTING OPERATIONS INVOLVES MANY REDUNDANT AND REPEATED PAPERWORK

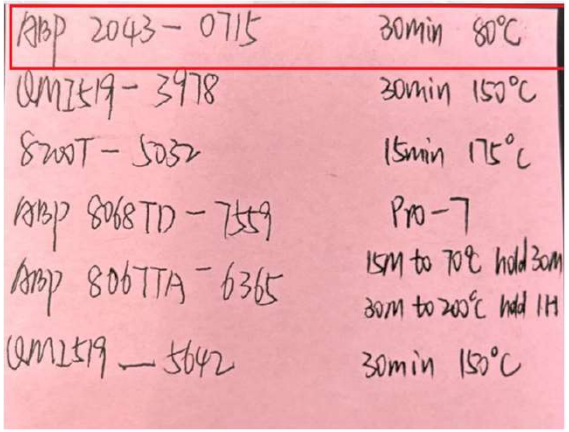
1. Sample registration on a **paper form**



2. Writing on w**hiteboard** to monitor test items and status

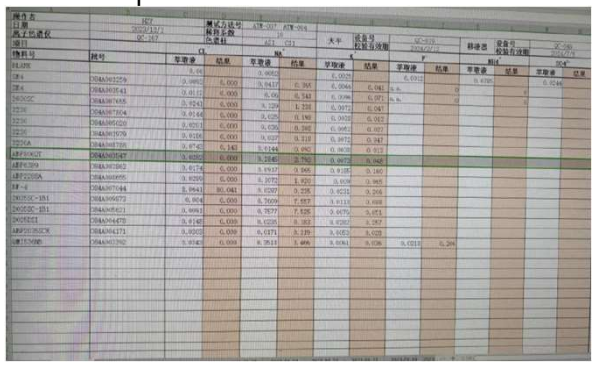


3. Writing on **sticky notes** to record curing & test parameters



4. Recording test results on **paper form**

5. Using **Excel sheets** for data inputs and calculations



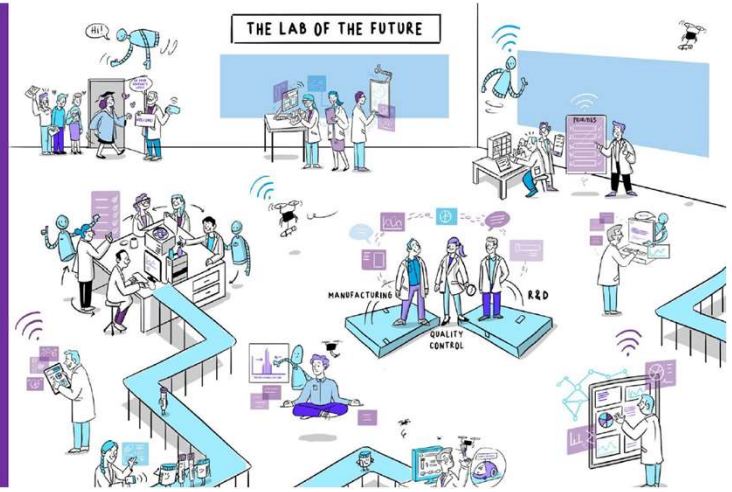
These examples are from the same QC lab.

QUALITY CONTROL EXCELLENCE PROGRAM TO STANDARDIZE AND SIMPLIFY

Current View of Laboratory of today…

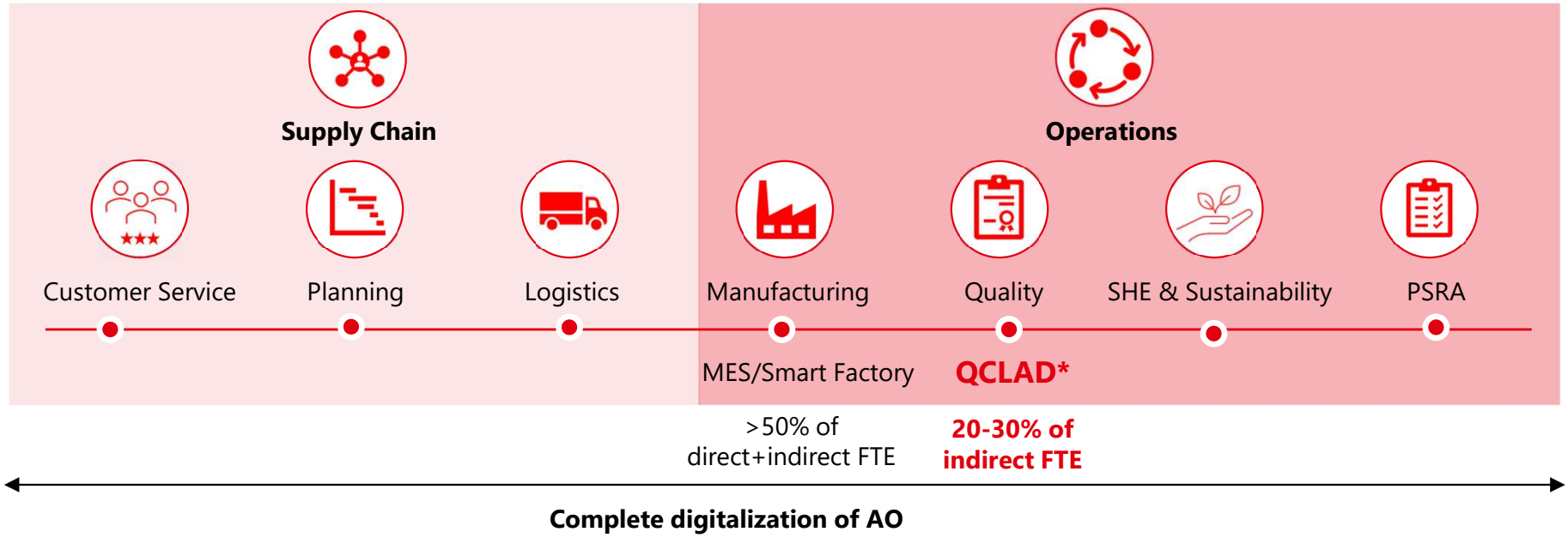
- Repeated and **redundant paperwork**
- **Non-value-added time** spent on manual processes
- **Lack of visibility** of important QC metrics
- Unable to utilize advanced data analytics
- **Messy flows and lab layout** results in movement waste
- **Non optimized** test plans and test methods

Together we can transform labs of today to labs fit for future of innovation



- Streamlined **digital information** flow
- QC team spends more time on **value-added activities**
- **Improved visibility** of important QC metrics
- **Enable advanced data analytics** for predictive quality
- **Leaner flows** and **lab layout**
- **Optimized tests** with **best available test methods**

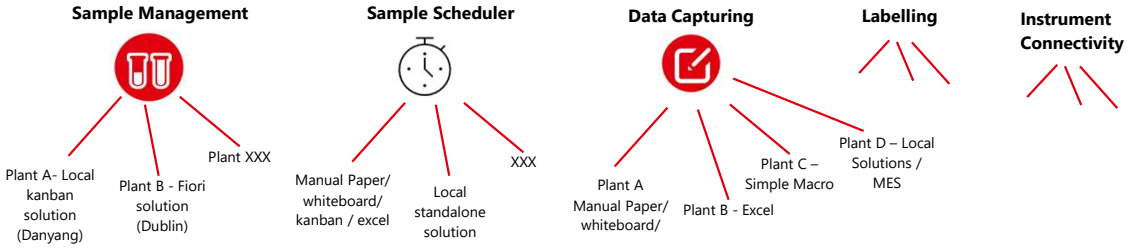
LAB DIGITIZATION IS KEY TO AO DIGITALIZATION AND EFFICIENCY IMPROVEMENTS



*QCLAD – QUALTY CONTROL LAB DIGITIZATION

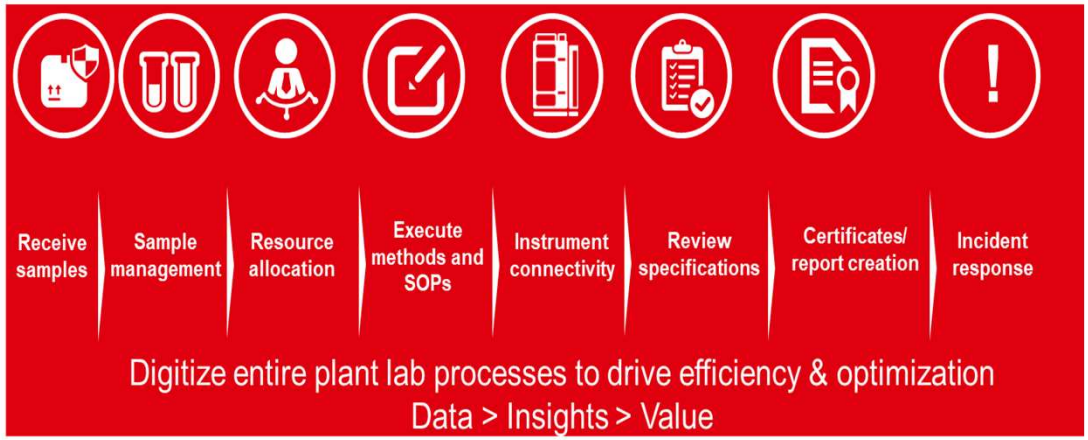
WAY FORWARD OPTIONS

OPTION1
Dispersed Local Solutions for Various Improvement Opportunities



- Do we need many solutions for same needs?
- How to integrate critical data for further analytics?
- Do we allow non standardized ways of work?
- Cost & Time spent on developing/maintaining different solutions?

OPTION2
Global platform with capability to scale based on site scenarios/ capabilities



Archetype 1 Solution - SAP (6 capabilities)

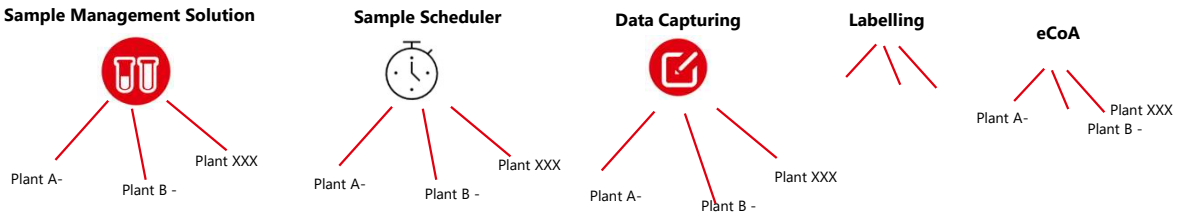
Archetype 2 Solution – SAP +MES (10 capabilities)

Archetype 3 Solution - SAP +MES+ OSDP (>12 capabilities)

- How to be applicable based on site needs?
- How to differentiate costs based on capabilities activated?
- Local speed vs standardized framework?

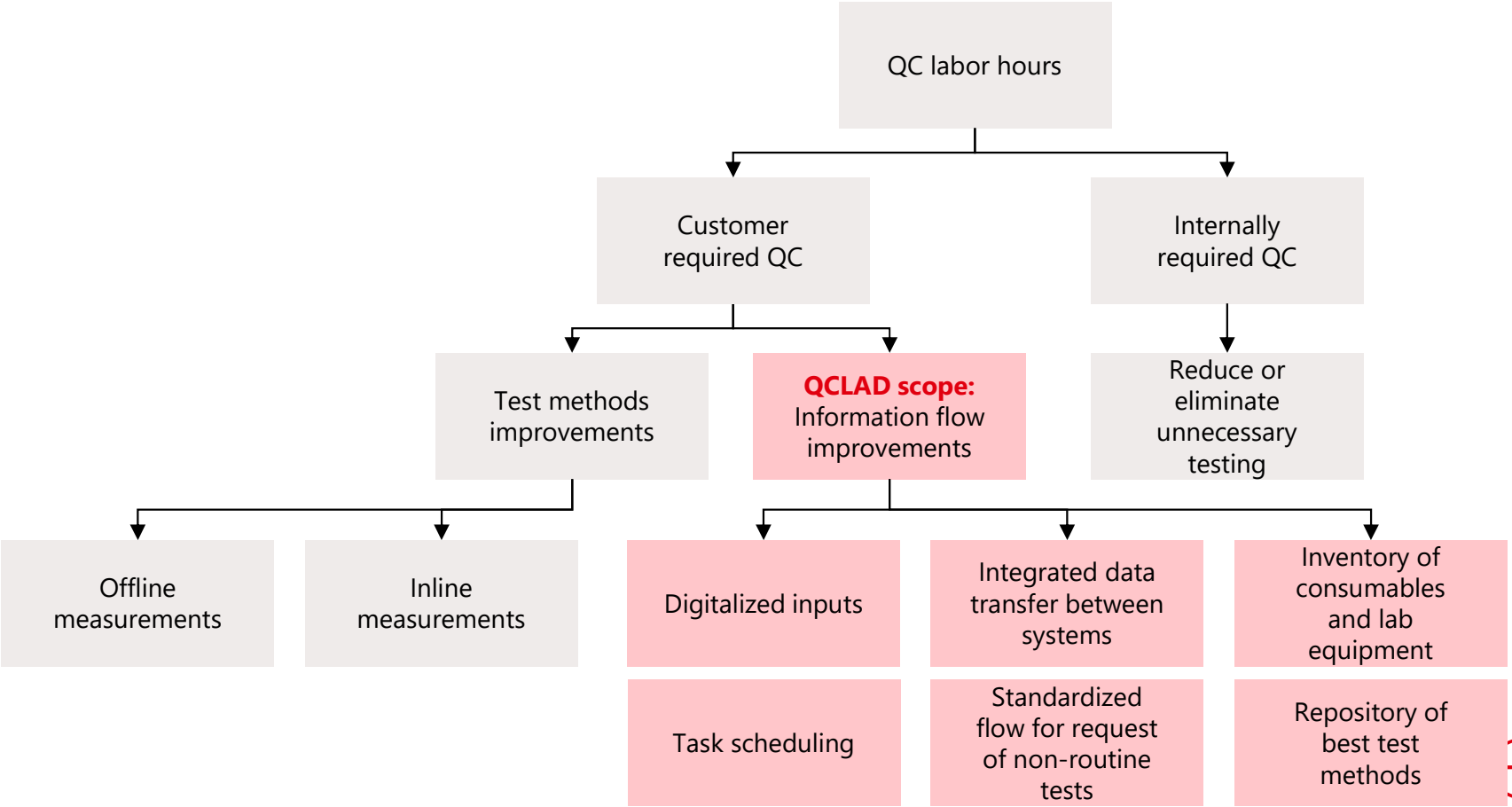
WAY FORWARD OPTIONS

OPTION3
Standard solutions for various capabilities that are not integrated

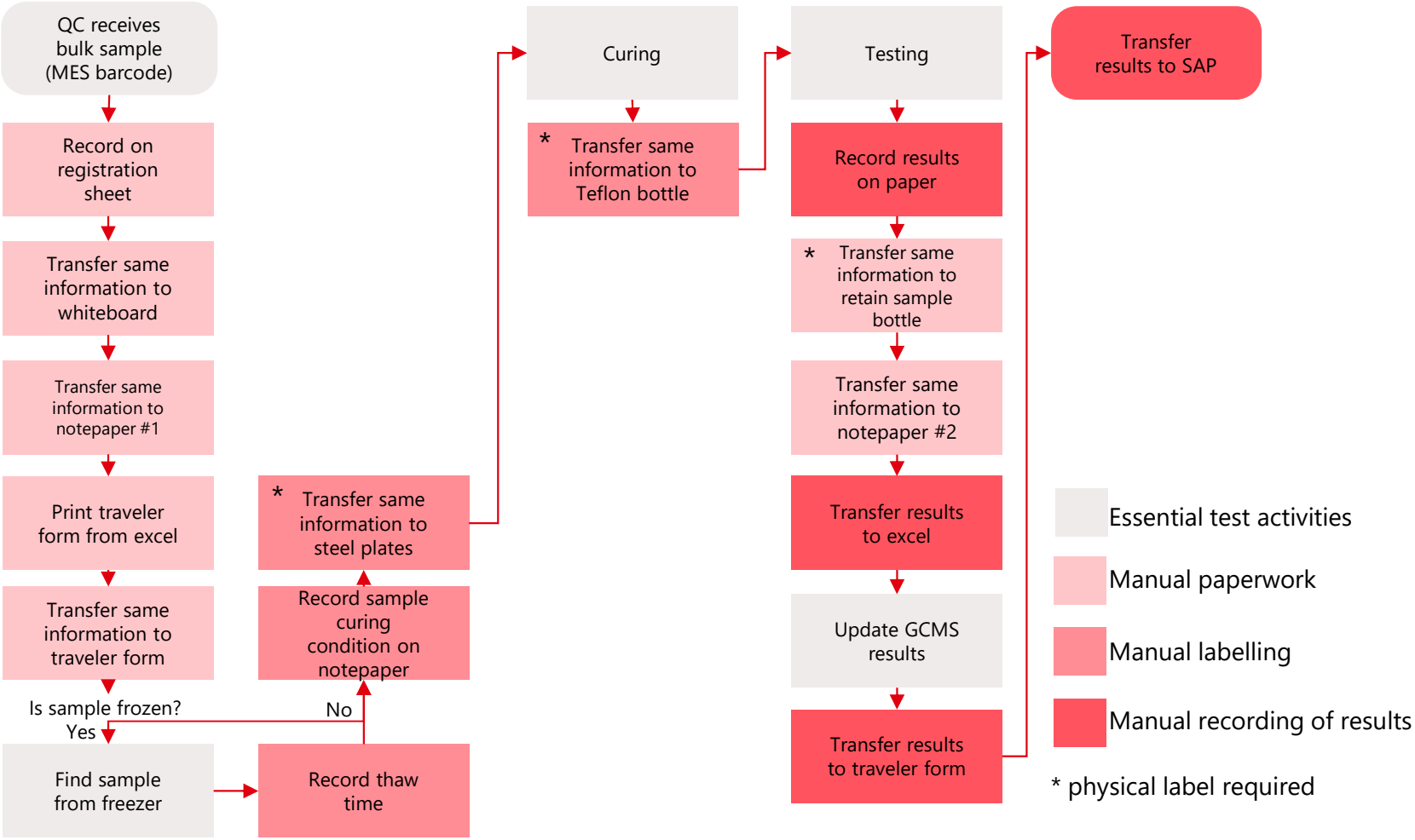


- Do how we deploy standard 9-12 capabilities required across 115 plants in AO for digitizing QC ?
- How to integrate critical data for further analytics?
- Do we allow non standardized ways of work?
- Cost & Time spent on developing/maintaining different solutions?

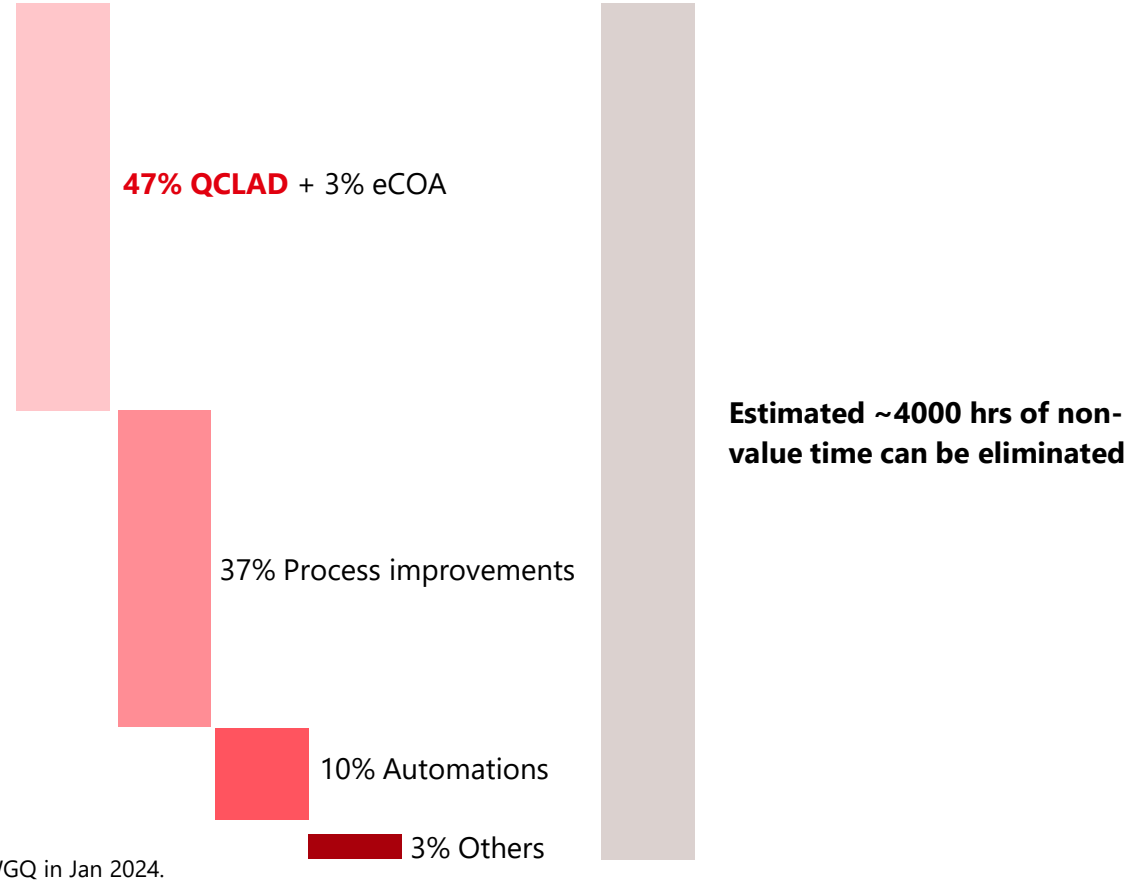
DIGITALIZATION IMPROVES QC EFFICIENCY THROUGH STREAMLINED DIGITAL INFORMATION FLOW



MAJORITY OF TEST ACTIVITIES ARE RELATED TO NON-VALUE ADDED REPEATED MANUAL WORK



MAJORITY OF NON-VALUE ADDED TIME CAN BE ELIMINATED THROUGH QCLAD



18 Data from QC Excellence workshop at WGQ in Jan 2024.

CYCLE TIME ANALYSIS SHOWS 28% TO 40% OF QC TEST TIME ARE SPENT ON MANUAL DATA ACTIVITIES

IONIC TEST TIME
20.1 HOURS

- ACTIVE MANPOWER HOURS
  8.0 HOURS (1 FTE)
  - TEST ACTIVITIES
    4.8h
    60%
  - **MANUAL DATA ACTIVITIES**
    3.2h
    40%
    - LABELLING
      0.6h
      8%
    - RECORDING RESULTS
      1.3h
      15%
    - INFORMATION FLOW
      1.4h
      17%
- WAITING TIME
  12.1 HOURS

DIE SHEAR STRESS TEST TIME
9.1 HOURS

- ACTIVE MANPOWER HOURS
  4.0 HOURS (0.5 FTE)
  - TEST ACTIVITIES
    2.9h
    72%
  - **MANUAL DATA ACTIVITIES**
    1.1h
    28%
    - LABELLING
      0.2h
      5%
    - RECORDING RESULTS
      0.7h
      18%
    - INFORMATION FLOW
      0.2h
      5%
- WAITING TIME
  5.1 HOURS

19 Time analysis done at WGQ during the Q Excellence workshop.

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DAY 1 VALUES

Use cases directly linked to functionality of solution and can begin contributing to productivity once system integrity is established, e.g., automatic data transfer and paperless working

Streamlined information flow
-5 mins per sample

Reduced repeated and redundant manual recordings of information.

Reduce results recording
-5 mins per sample

Results directly pushed from equipment and digital tablets to systems that are integrated.

Automated printing of labels
-3 mins per sample

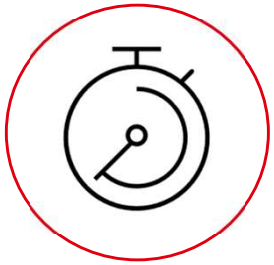
Reduce time spent to manually write sample labels for identifications.

Reduce lab consumables inventory
-10% of inventory cost

Increase visibility and management of lab consumables to reduce overall inventory cost.

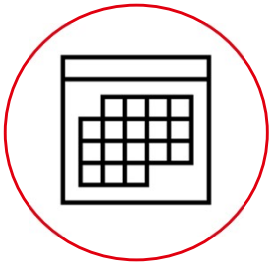
DAY 2 VALUES

Additional productivity opportunities driven by continuous improvement enabled by data visualization and analysis.



QC test cycle time reduction
-10% cycle time

Reduce long cycle time test processes by using data to improve test methods and equipment through discussions with PD.



Improve planning and scheduling
+10% QC utilization

Optimize planning and scheduling based on accurate QC labor times, test cycle times and production requirements.

Sharing of QC lab best practices
-10% cycle time

Data enable performance monitoring and benchmarking of optimum test methods and processes regionally and globally.

Data-driven CI
+25% accelerated improvement

Increased & accelerated value creation from continuous improvement using advanced data analytics

SCALABLE CAPABILITIES BASED ON SITES

draft

Archetype1 - Site Capability

- Site with just SAP
- 90% of lab equipment are TYPE C& D
- Labelling capabilities exists (Vegam/SAP)

QCLAD VALUE CREATION

- Sample Manager (for lifecycle of sample management)
- Manual Test Scheduling
- Non routine tests management
- LES (QC tests / workflows / SOP management)
- Interface with SAP QM
- Basic Consumable Management
- Basic reports and dashboards
- User access management

COST IMPACT

- License cost for X licenses (based on QC operators)
- *One time integration cost if labelling is via SAP for the initial plant

Archetype2 - Site Capability

- Site with SAP
- 25 % of lab equipment's are TYPE A&B
- Labelling capabilities exist (Vegam /SAP)
- MES or Vegam QM capability exists

QCLAD VALUE CREATION

- Sample Manager (for lifecycle of sample management)
- Manual Test Scheduling
- Non routine tests management
- LES (QC tests / workflows / SOP management)
- Interface with SAP QM, MES or Vegam
- Lab Management
- Data capture from instrument
- Reporting and Dashboard
- User access management

COST IMPACT

- License cost for X licenses (based on QC operators)
- *One time integration cost if labelling is via SAP)
- *One time integration cost with MES Wonderware for the initial plant.

Archetype3- Site Capability

- Site with SAP
- >25 % of lab equipment's are TYPE A&B
- Labelling capabilities exist (Vegam /SAP)
- MES or Vegam QM capability exists
- ODSP solution deployed

QCLAD VALUE CREATION

- Sample Manager (for lifecycle of sample management)
- Sampling Scheduler
- Non routine tests management
- LES (QC tests / workflows / SOP management)
- Interface with SAP QM, MES or Vegam
- Lab Management
- Data capture from instrument
- Reporting and Dashboard
- User access management

COST IMPACT

- License cost for X licenses (based on QC operators)
- *One time integration cost if labelling is via SAP)
- *One time integration cost with MES Wonderware.

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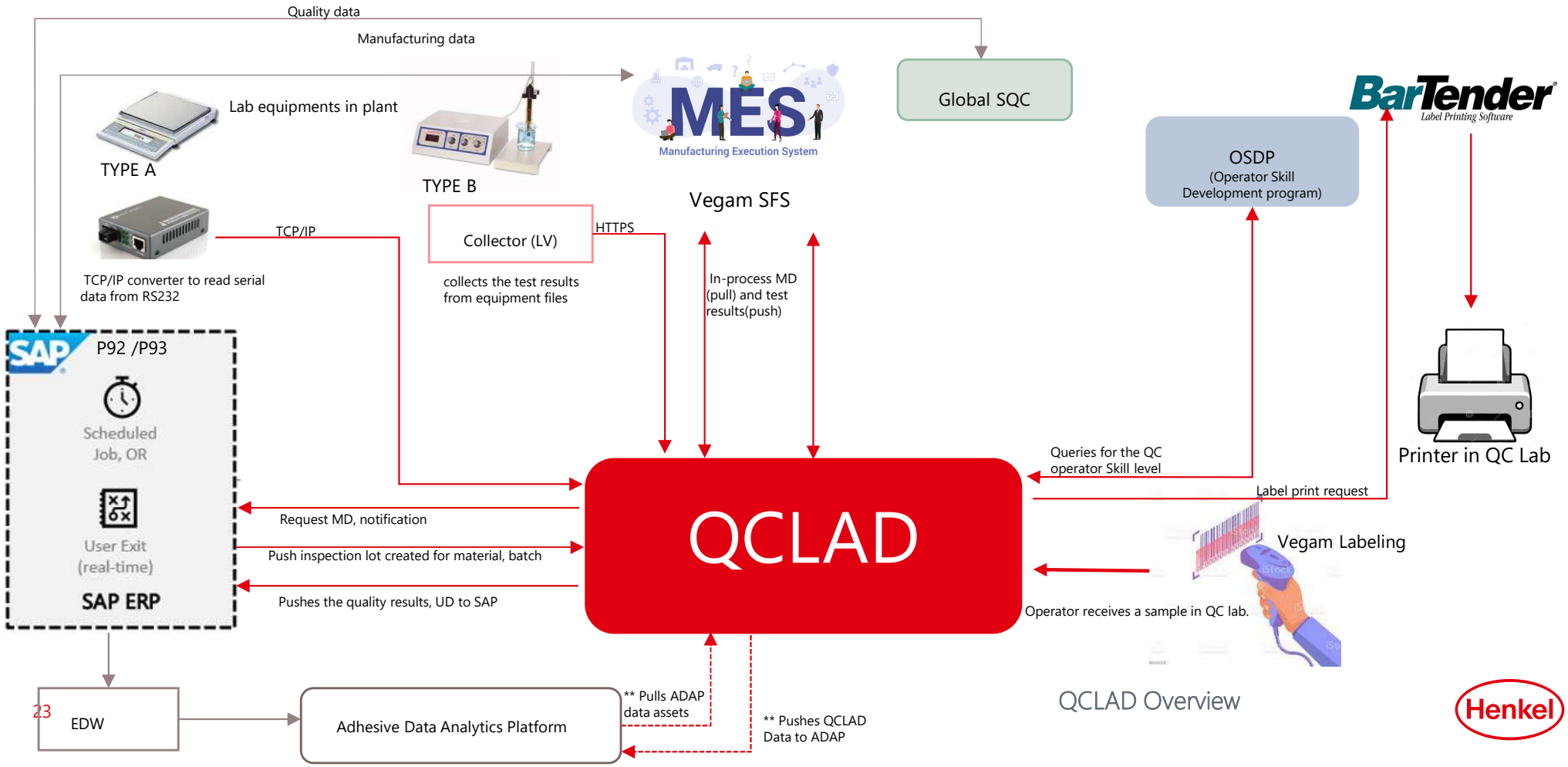
*One time integration cost needs to be adhered in project cost

OVERVIEW OF ARCHITECTURE – ARCHETYPE 3



EQUIPMENT TYPES

Type A:
Instruments with interface

- Such as balance, moisture tester Titrator etc.

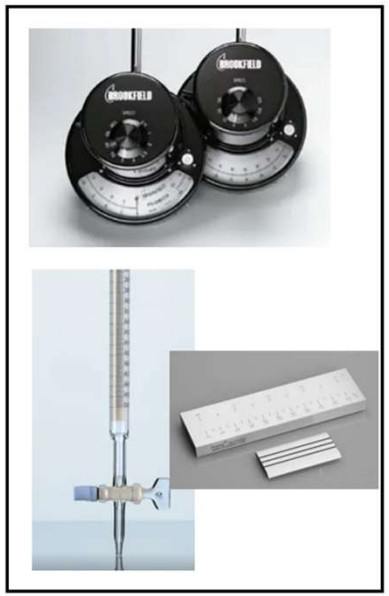
Type B:
Instruments with Software

- Such as Rheometer, tensile machine etc.

Type C:
Instruments Image Identification

- Such as oven, pH meter water bath etc.

Type D:
Instruments with scale etc.



- Such as dial plate, scale need to read by eyes etc.

THANK YOU